

## [WISER Global Quantum+AI Program 2026](http://www.thewiser.org/summer-program-2026)

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### **WISER <> Moderna**

## Optimization of mRNA Secondary Structure Prediction Using Quantum Computing

### About Moderna:

Moderna is a biotechnology company pioneering messenger RNA, or mRNA, medicines. Moderna's platform focuses on using mRNA to instruct cells to produce proteins that may help prevent or treat disease. This challenge explores a computational problem relevant to mRNA design: predicting and optimizing RNA secondary structure.

### Use Case Description:

The prediction of mRNA secondary structures plays an important role in the design and analysis of mRNA-based medicines. Secondary structure can influence molecular stability, translation efficiency, and manufacturability. From a computational perspective, predicting RNA secondary structure is challenging because the number of possible folds grows rapidly with sequence length. Classical tools such as ViennaRNA use thermodynamic models and dynamic programming-based methods to identify Minimum Free Energy, or MFE, structures. However, the broader folding landscape remains difficult to explore exhaustively, especially when richer structural constraints are considered. This challenge asks participants to investigate whether quantum or quantum-inspired optimization methods can be used to formulate and explore the mRNA secondary-structure prediction problem. The goal is not necessarily to outperform classical methods during the program, but to reproduce known benchmark structures for small RNA sequences and analyze how quantum resource requirements scale with sequence length.

### Problem Statement:

Participants will develop a quantum or quantum-inspired algorithm for mRNA secondary-structure prediction, with a focus on MFE-based folding. Given an input RNA sequence consisting of the nucleotides A, U, C, and G, participants should formulate the space of possible secondary structures as an optimization problem and use a quantum computing approach to search for low-energy candidate structures. Participants may use classical RNA folding software, such as the ViennaRNA Python package, to generate reference structures and evaluate energies. Candidate solutions should be compared against the MFE structure returned by classical tools.

### Challenge Duration:

- Teams start working in July 2026

- Submission on **August.07, 2026**. There will be no extension to the submission deadline.

### Team Guidelines:

- Max. Team size - **3 participants per team**.
- Ensure you are enrolled as a participant in the WISER Quantum 2026 program.
- Everyone is eligible to participate in this challenge and win awards.
- Best participants are invited directly to interview for WISER research fellowships.

### Challenge Tasks/ Deliverables:

1. Background review: Review introductory resources on RNA secondary structure, Minimum Free Energy folding, dot-bracket notation, and quantum optimization methods. Prepare a short summary of the biological problem, the computational challenge, and the proposed quantum approach.
2. Classical benchmark generation: Use the ViennaRNA Python package to generate reference MFE structures for short RNA sequences. Participants may use randomly generated RNA sequences or provided example sequences.

Python

```
import RNA
sequence = "GGAGCAAACUUGUCGAUUGAGAACAAAUAACAGAAUUUGCUUG"
structure, mfe = RNA.fold(sequence)
print(structure, mfe)
```

3. Energy evaluation of candidate structures: Represent candidate RNA secondary structures using dot-bracket notation and evaluate their energies using ViennaRNA. The energy gap between a candidate structure and the MFE structure should be used as one benchmark metric.

Python

```
structure = ".....((((.....)))....."
energy = RNA.fold_compound(sequence).eval_structure(structure)
print(energy)
```

4. Quantum or quantum-inspired formulation: Formulate RNA folding as an optimization problem suitable for a quantum or quantum-inspired algorithm. Possible approaches may include QAOA, VQE, quantum annealing, Grover-style search, tensor-network-inspired methods, or other suitable optimization strategies.
5. Implementation and benchmarking: Implement the proposed approach for small RNA sequences and compare the predicted structures or energies against the ViennaRNA reference results. Report accuracy, energy gap from the reference MFE structure, runtime, and observed limitations.

6. Scaling and quantum resource analysis: Analyze how the method scales with RNA sequence length. Include estimated qubit count, circuit depth, number of variables, runtime, and practical limitations on near-term quantum devices or simulators.
7. Final submission package:
  - Submit code, documentation, results, and a short presentation explaining the approach, findings, and future directions.
  - The submission should be easy to access and reproducible.
8. *Optional Advanced Tasks*:
  - Explore pseudoknot-aware formulations or explicitly state why pseudoknots are excluded from the initial model.
  - Compare multiple quantum encodings of the same folding problem and discuss tradeoffs in qubit count and constraint enforcement.
  - Evaluate algorithm robustness under sampling noise or hardware-inspired noise models.

#### Quantum Hardware/ Platform:

- Participants may use any quantum SDK or platform of their choice, including Qiskit, PennyLane, Cirq, Amazon Braket, IBM Quantum, D-Wave, or local simulators.
- Use of real quantum hardware is optional. Simulations are sufficient for completing the challenge.
- For noise-aware extensions, participants may use any available hardware noise model or simulator backend.

#### Judging Criteria:

- Clarity and correctness of the problem formulation.
- Quality of the quantum or quantum-inspired algorithm design.
- Ability to reproduce or approximate classical MFE benchmark structures.
- Quality of benchmarking against ViennaRNA reference results.
- Depth of scaling and quantum resource analysis.
- Technical merit, creativity, and novelty.
- Code quality, documentation, and reproducibility.
- Communication and presentation quality.

#### Resources:

- ViennaRNA Python package: <https://viennarna-python.readthedocs.io/en/master/>
- ViennaRNA dot-bracket notation documentation:  
[https://www.tbi.univie.ac.at/RNA/ViennaRNA/refman/io/rna\\_structures.html#dot-bracket-notation](https://www.tbi.univie.ac.at/RNA/ViennaRNA/refman/io/rna_structures.html#dot-bracket-notation)
- High-level introductory deck: [https://alexgalda.github.io/quantum\\_mRNA\\_optimization/](https://alexgalda.github.io/quantum_mRNA_optimization/)
- Useful reading: introductory materials on RNA secondary structure and Minimum Free Energy folding, QAOA, VQE, quantum annealing, and quantum-inspired optimization.

**Data Privacy/ Eligibility Restriction:**

Participants should use only publicly available, synthetic, or randomly generated RNA sequences. No confidential Moderna data, patient data, clinical data, proprietary sequence data, or personally identifiable information should be used. This challenge is intended as an educational and research exercise.